Pulses in Cables Lab Report

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1 Introduction

The purpose of this set of experiments is to investigate the propagation of electrical pulses along transmission lines and to determine the characteristic impedance and signal velocity in both a discrete LC transmission-line model and coaxial cables. These effects arise because real circuits do not transmit signals instantaneously; instead, voltage and current propagate as electromagnetic waves along conductors with distributed inductance and capacitance [1]. In these experiments, a waveform generator is used to send short pulses through a transmission line while an oscilloscope records both the outgoing and returning signals. The delay between pulses is used to determine the propagation speed along the line, which was determined to be 7530 ± 40 m/s, which is within 8% of the predicted value. Measurements are also performed using coaxial cables to determine the speed of travel in the coaxial cables, which was $(2.084 \pm 0.009) \cdot 10^8$ m/s in the 48.36 metre long cable, and $(2.028 \pm 0.005) \cdot 10^8$ m/s in the 75.84 metre long cable. The attenuation factors were also measured, which were determined to be -0.047 ± 0.002 dB/m in the 48.36 metre long cable, and -0.044 ± 0.002 dB/m in the 75.84 metre long cable.

2 Equipment and Methodology

2.1 Equipment

The experimental setup consisted of a waveform generator, digital oscilloscope, LC transmission-line, a coaxial cable, and a series of resistive terminals. The transmission line had 41 individual inductances, each at 1.5 mH $\pm$10%, and 41 individual capacitances, each at 0.01μF $\pm$3%. The coaxial cable had a length of 75.84 ± 0.02 metres. The oscilloscope has a stated voltage uncertainty of $\pm$3% [2]. The time uncertainty was chosen to be $\pm$1ns, since the stated uncertainty was significantly smaller than the smallest measurement increment using cursors, which was around 1 ns. [2].

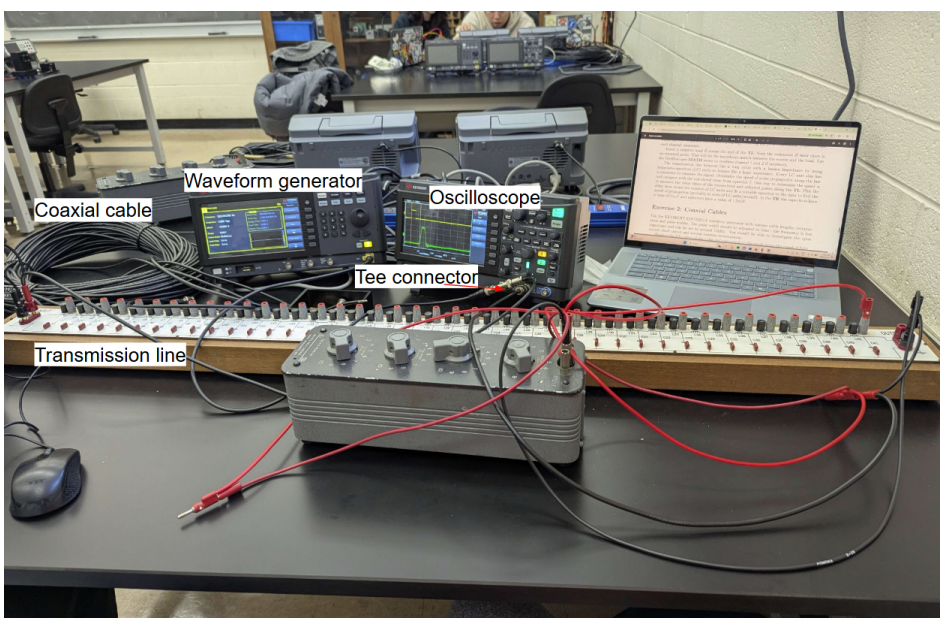

Figure 1: Our labeled experimental setup. At the suggestion of the practicals coordinator, we used the fixed value resistive terminations rather than the variable resistor in this photo.

2.2 Methodology

2.2.1 The Impedance of a Transmission Line

The waveform generator was used to produce a square wave of 300 Hz with a width of 75μs through a T-connector and then through the transmission line. One branch of a T-connector was connected to Channel 1 of the digital oscilloscope to monitor the incident pulse, while the input to the transmission line was connected to the second branch. The output of the transmission line was connected to Channel 2 of the oscilloscope to observe transmitted and reflected pulses.

3 Results and Discussion

3.1 The Impedance of a Transmission Line

As the transmission setup is composed of LC-units, each unit consists of a capacitor with a capacitance of 0.01μF and an inductance of 1.5 mH. These LC units are spaced somewhat evenly; however, there is some variability within the spacing between LC units larger than the uncertainty of the meter stick used to measure the distances between them. Thus, two sets of data will be presented for speed: the first will be to determine speed in terms of LC units, and the second will determine speed in m/s.

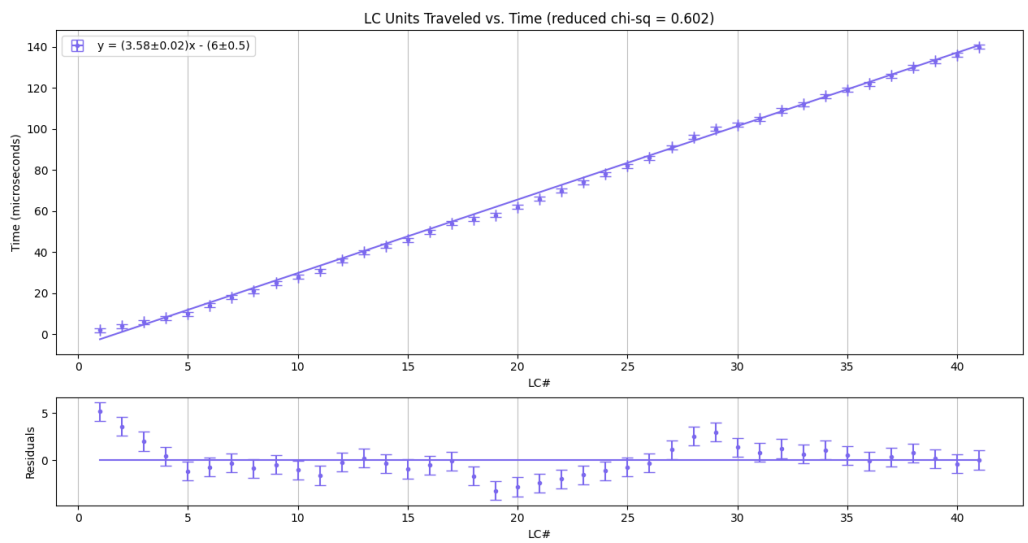

Figure 2: A graph showing the relationship between the time elapsed (in microseconds) and the number of LC units traveled. The reduced chi squared value is 0.602, implying a slight amount of overfitting.

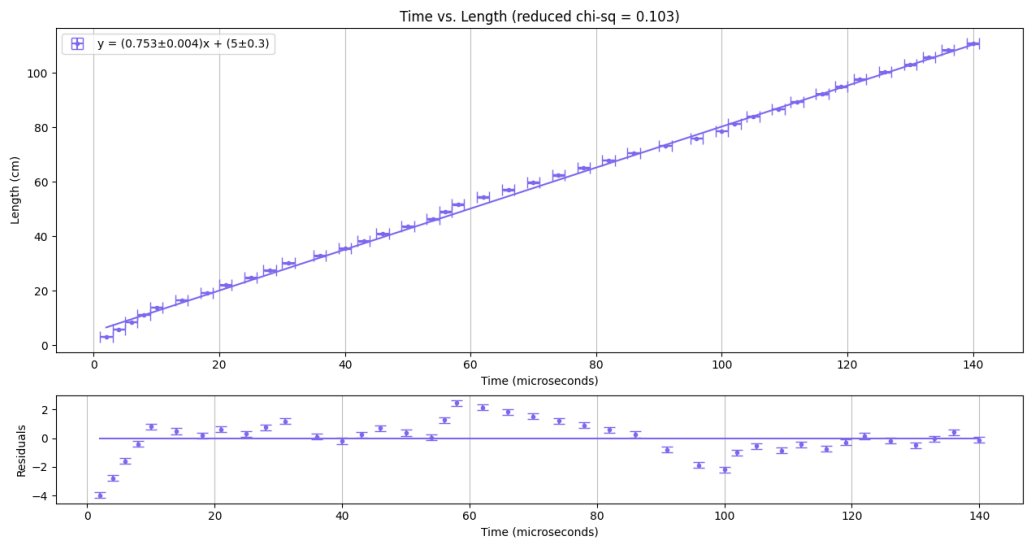

Figure 3: A graph showing the relationship between the time elapsed (in microseconds) and the distance traveled (in centimeters). The reduced chi squared value is 0.103, implying a potentially significant amount of overfitting.

From Figure 3, we can see the measured speed of the pulse traveling through the transmission line is 0.753 cm/μs, or 7530 m/s. From the slope uncertainty, we get the uncertainty in speed to be $\pm$ 40 m/s. The average spacing between subsequent LC units is 2.7 cm, so the theoretical velocity of pulse travel is:

$$v = \frac{1}{\sqrt{L_0 C_0}} \tag{1}$$

$$L_0 = \frac{1.5 \cdot 10^{-3}}{0.027} \text{H/m} \tag{2}$$

$$C_0 = \frac{0.01 \cdot 10^{-6}}{0.027} \text{F/m} \tag{3}$$

$$v = \sqrt{\frac{(0.027)^2}{(1.5 \cdot 10^{-3})(0.01 * 10^{-8})}} \tag{4}$$

$$v = 6971 \text{ m/s} \tag{5}$$

Thus, our experimental result is in somewhat good agreement with the theoretical value. While the theoretically predicted value is not within the uncertainty of the experimental one, our experimental value is within 8% of the theoretically predicted one. Additionally, we calculated the reduced chi-squared to have a value of 0.103, which implies overfitting, contributing to the disparity.

The oscilloscope can be ignored when considering reflections because its input impedance (1 MΩ) [2] is much larger than the provided impedance of the transmission line (50 Ω)[1]. As a result, a negligible current flows into the oscilloscope, so it does not significantly affect the impedance matching or reflections in the transmission line.

3.2 Speed of Travel in a Coaxial Cable

We measured the speed of travel of a pulse in two coaxial cables of different lengths, with a variety of resistive terminations as well. Our data for both are listed in the table below:

Cable Length (m)	Termination	Time (±1 ns)
48.36 ± 0.02	Open	232
	Shorted	238
	10 Ω	238
	27 Ω	240
	51 Ω	242
	100 Ω	246
	270 Ω	246
75.84 ± 0.02	Open	374
	Shorted	374
	10 Ω	378
	27 Ω	380
	51 Ω	380
	100 Ω	384
	270 Ω	384

Table 1: A table showing the travel time of a pulse through two cables of lengths 48.36 ± 0.02 m and 75.84 ± 0.02 m, with a variety of resistive terminations.

The speed of pulse propagation can be calculated simply by dividing the cable length by the travel time for the open circuit. The open-circuit termination produces complete reflection without inversion, making the reflected pulse easier to identify on the oscilloscope, and therefore the delay to be easier to measure. As the reflection coefficient is r=1, the reflected pulse has the same polarity as the incident pulse.

For the cable of length 48.36 ± 0.02 m, this speed is:

$$v = \frac{48.36}{(232 \cdot 10^{-9})} = (2.084 \pm 0.009) \cdot 10^8 \text{ m/s} \tag{6}$$

For the cable of length 75.84 ± 0.02 m, this speed is:

$$v = \frac{75.84}{(374 \cdot 10^{-9})} = (2.028 \pm 0.005) \cdot 10^8 \text{ m/s} \tag{7}$$

We can compare this with the provided values from the lab manual. Using the following equation, and the speed of light ($3 \cdot 10^8$ m/s) [3].we can get the predicted speed of travel.

$$v = \frac{c}{\sqrt{\epsilon \cdot \mu}} = \frac{3 \cdot 10^8}{\sqrt{2.25}} = 2 \cdot 10^8 \text{ m/s} \tag{8}$$

Thus, we have reasonably close agreement between the predicted and actual values of the speed of travel in the coaxial cable.

3.3 Attenuation Factor and Impedance of a Coaxial Cable

For each combination of cable length and resistive termination used in the prior section, we also measured the peak voltage of the input waveform and the peak voltage of the reflected waveform, the results of which can be seen in the table below:

Cable Length (m)	Termination	Input Peak Voltage (V ± 3%)	Output Peak Voltage (V ± 3%)
48.36 ± 0.02	Open	4.35	3.35
	Shorted	4.31	-3.56
	10 Ω	4.3	-2.25
	27 Ω	4.33	-1.175
	51 Ω	4.25	0
	100 Ω	4.3	1.00
	270 Ω	4.35	2.18
75.84 ± 0.02	Open	4.2	2.85
	Shorted	4.2	-3.25
	10 Ω	4.2	-1.95
	27 Ω	4.2	-1.05
	51 Ω	4.2	0
	100 Ω	4.2	0.85
	270 Ω	4.2	1.83

Table 2: A table showing the travel time of a pulse through two cables with a variety of resistive terminations.

From the table, we can calculate the attenuation factor of the cables by using the formula provided [1]

For the 48.36 ± 0.02 m long cable, $\frac{V_{reflected}}{V_{initial}} = \frac{3.35}{4.35} = 0.77$, so the attenuation factor is:

$$\text{Attenuation} = \frac{10\log_{10}(0.77^2)}{48.36} = -0.047 \pm 0.002 \text{ dB/m} \tag{9}$$

Meanwhile, for the 75.84 ± 0.02 m long cable, $\frac{V_{reflected}}{V_{initial}} = \frac{2.85}{4.2} = 0.68$, so the attenuation factor is:

$$\text{Attenuation} = \frac{10\log_{10}(0.68^2)}{75.84} = -0.044 \pm 0.002 \text{ dB/m} \tag{10}$$

Additionally, we can find the value of Z_0 for both cables, as that is simply the value for which the output peak voltage is zero. Consulting the table, we see that the value of Z_0 for both cables is around 51 Ω, which matches with the reference value of 50 Ω reported in the lab manual. As our measurements were discrete with large intervals between resistances, we cannot report the value of Z_0 to any significant numerical accuracy.

4 Uncertainty Analysis

The uncertainty of the slopes and intercepts of the graphs shown in subsection 3.1 was calculated using the formula from [4]:

$$\Delta = N\sum x_i^2 - (\sum x_i)^2 \tag{11}$$

$$s_{y,x}^2 = \frac{1}{N-2}\sum[y_i - (mx_i + b)]^2 \tag{12}$$

$$s_m = \sqrt{N\frac{s_{y,x}^2}{\Delta}} = \sqrt{\frac{N}{N-2}\frac{\sum[y_i - (mx_i+b)]^2}{N\sum x_i^2 - (\sum x_i)^2}} \tag{13}$$

$$s_b = \sqrt{\frac{s_{y,x}^2\sum x_i^2}{\Delta}} = \sqrt{\frac{1}{(N-2)}\frac{\sum[y_i - (mx_i+b)]^2\sum x_i^2}{N\sum x_i^2 - (\sum x_i)^2}} \tag{14}$$

An example calculation is shown below for the relationship between time and distance traveled for the transmission line, where s_m is the slope uncertainty and s_b is the uncertainty in the y intercept (see Appendix

A for the data used for the calculations):

$$s_m = \sqrt{\frac{41}{39}\frac{[(3.1-(0.753\cdot 2+5)]^2+\cdots+[(110.9-(0.753\cdot 140+5)]^2}{41(2^2+4^2+\cdots+140^2)-(2+4+\cdots+140)^2}} \tag{15}$$

$$s_m = 0.004 \tag{16}$$

$$s_b = \sqrt{\frac{1}{(39)}\frac{([(3.1-(0.753\cdot 2+5)]^2+\cdots+[(110.9-(0.753\cdot 140+5)]^2)(2^2+4^2+\cdots+140^2)}{41(2^2+4^2+\cdots+140^2)-(2+4+\cdots+140)^2}} \tag{17}$$

$$s_b = 0.3 \tag{18}$$

Uncertainties for v were calculated using the formula from [4]:

$$\frac{\Delta z}{z} = \sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta y}{y}\right)^2} \tag{19}$$

An example calculation is shown below for the uncertainty in speed of travel in the coaxial cable.

$$\frac{\Delta v}{v} = \sqrt{\left(\frac{1}{232}\right)^2 + \left(\frac{0.02}{48.36}\right)^2} \tag{20}$$

$$\Delta v = (0.00433)(2.084\cdot 10^8) \tag{21}$$

$$\Delta v = 0.009 \tag{22}$$

The uncertainty for attenuation factor was found by plugging in the percent uncertainties of the voltages (3%) into the equation (the uncertainty in length is negligible compared to the uncertainty in voltage). and then multiplying that value (4.2%) by the attenuation factor [4].

4.1 Sources of error and uncertainty

We had several sources of error in this experiment, which we believe should be covered by the uncertainties provided as well as our additional measurement uncertainties. The largest uncertainty arose from oscilloscope timing measurements. Delay times were sensitive to cursor placement and screen resolution. Although the actual oscilloscope measurement based uncertainty was very small [2], the practical uncertainty in delay measurements is estimated to be approximately around 1% of the measured value, as that was the approximate amount of granularity we had with the cursors. The finite rise times of real pulses meant that the pulses were broadened and slightly distorted from the ideal shapes, which meant that we had to approximate the location of beginning of each pulse.

Voltage measurements used to determine attenuation and reflection amplitudes were limited by the oscilloscope vertical accuracy (± 3%) and by noise in the signal. Small variations in pulse amplitude made it difficult to identify the exact peak value of reflected signals. Because the experimental line is composed of discrete elements rather than the continuous transmission line outlined in the model, the real line probably will not reflect the idealized model, although this effect should be small.

References

[1] Lee, C., Vahabi, T., Serbanescu, R., Helt, L., and Edwards, N. *Pulses in Cables*. Department of Physics, University of Toronto, March 2025. Revision 3fcb776.

[2] Keysight Technologies. *InfiniiVision 1000 X-Series Oscilloscopes Data Sheet*. Keysight Technologies, 2023.

[3] National Institute of Standards and Technology. Speed of light in vacuum, 2022. CODATA recommended values of the fundamental physical constants. Accessed: 2026-02-17.

[4] David M. Harrison. *Error Analysis in Experimental Physical Science*. Department of Physics, University of Toronto, 2011. Last revision: September 10, 2011.

Appendix A: Raw Data Tables

Transmission Line - LC Units vs. Time

Time (microseconds)	L#	Xerr	Yerr
2	1	1	0
4	2	1	0
6	3	1	0
8	4	1	0
10	5	1	0
14	6	1	0
18	7	1	0
21	8	1	0
25	9	1	0
28	10	1	0
31	11	1	0
36	12	1	0
40	13	1	0
43	14	1	0
46	15	1	0
50	16	1	0
54	17	1	0
56	18	1	0
58	19	1	0
62	20	1	0
66	21	1	0
70	22	1	0
74	23	1	0
78	24	1	0
82	25	1	0
86	26	1	0
91	27	1	0
96	28	1	0
100	29	1	0
102	30	1	0
105	31	1	0
109	32	1	0
112	33	1	0
116	34	1	0
119	35	1	0
122	36	1	0
126	37	1	0
130	38	1	0
133	39	1	0
136	40	1	0
140	41	1	0

Transmission Line - Time vs. Length

Time (microseconds)	Length (cm)	Xerr	Yerr
2	3.1	1	0.2
4	5.8	1	0.2
6	8.5	1	0.2
8	11.2	1	0.2
10	13.9	1	0.2
14	16.6	1	0.2
18	19.3	1	0.2
21	22	1	0.2
25	24.7	1	0.2
28	27.4	1	0.2
31	30.1	1	0.2
36	32.8	1	0.2
40	35.5	1	0.2
43	38.2	1	0.2
46	40.9	1	0.2
50	43.6	1	0.2
54	46.3	1	0.2
56	49	1	0.2
58	51.7	1	0.2
62	54.4	1	0.2
66	57.1	1	0.2
70	59.8	1	0.2
74	62.5	1	0.2
78	65.2	1	0.2
82	67.9	1	0.2
86	70.6	1	0.2
91	73.3	1	0.2
96	76	1	0.2
100	78.7	1	0.2
102	81.4	1	0.2
105	84.1	1	0.2
109	86.8	1	0.2
112	89.5	1	0.2
116	92.2	1	0.2
119	94.9	1	0.2
122	97.6	1	0.2
126	100.3	1	0.2
130	103	1	0.2
133	105.7	1	0.2
136	108.4	1	0.2
140	110.9	1	0.2